

MELANOMA

AT A GLANCE

Rising incidence worldwide; U.S. estimated lifetime risk of invasive melanoma is 1 in 1500 if born in 1935, 1 in 600 if born in 1960, 1 in 62 if born in 2006 (1 in 34 with inclusion of in situ melanoma).

Risk factors: history of sunburns and/or heavy sun exposure, blue or green eyes, blonde or red hair, fair complexion, >100 typical nevi, any atypical nevi, prior personal or family history of melanoma, *p16* mutation.

Mean age of diagnosis: 52 years — 10 to 15 years younger than more common cancers (e.g., breast, lung, colon, prostate).

Most common locations: back in men; lower extremities, then trunk in women.

Clinical features for recognition (ABCDE criteria):

- **A** – Asymmetry
- **B** – Irregular borders
- **C** – Color variegation
- **D** – Diameter >6 mm (common interpretation), or "difference" — the "ugly duckling" sign, or change in size, shape, color, or persistent pruritus
- **E** – Evolving over time

Melanoma follows a highly variable course and is a heterogeneous disorder. It is surgically curable when diagnosed and treated early, but potentially lethal when diagnosed at a late stage.

Sunburn and Melanoma Risk

Sunburn history, particularly blistering and peeling burns, serves as a surrogate for intermittent intense sun exposure. A review of published literature shows a significant positive association between childhood sunburns and increased risk

of melanoma. **One blistering sunburn in childhood more than doubles a person's lifetime risk** of developing melanoma. Sunburns in adulthood also contribute to melanoma risk.

Data from the CDC's Behavioral Risk Factor Surveillance System indicate that:

- Nearly **32% of adults ≥ 18 years**
- **58% of adults aged 18–29 years**
- Over **40% of children**

report at least one sunburn per year (defined as red skin lasting >12 hours).

Crude sunburn prevalence is highest among white adults, with higher rates observed in Caucasian populations in the Midwest, possibly due to increased sun sensitivity or lower adherence to sun-safe behaviors.

Geographic and Migration Patterns

Migration studies show increased melanoma risk in individuals who: - Spent childhood in sunny regions - Emigrated to sunnier areas at a young age

Younger migrants have a higher risk compared to adult immigrants, and risk increases with **longer duration of residence** in sunny locales.

Melanoma incidence and mortality among Caucasians show a **latitude gradient**: rates are higher closer to the equator and decrease with increasing latitude, correlating with UV radiation dose. This pattern is observed in geographically diverse regions including the United States, New Zealand, and Australia. However, this gradient may be influenced by other risk factors such as skin phenotype and behavioral sun exposure.

TABLE 124-1 Risk Factors for Cutaneous Melanoma

- **Ultraviolet radiation exposure**
- Blistering sunburns at any age

- Intermittent or sporadic high-intensity sun exposure
- Excessive chronic sunlight exposure
- **Phenotypic characteristics**
- Fair skin, inability to tan, tendency to sunburn or freckle (Skin Phototype I and II)
- Blue or green eyes
- Red or blond hair
- Numerous typical nevi (>100) and/or presence of atypical nevi
- Large congenital nevus
- **Personal or family history**
- Prior melanoma
- Family history of melanoma
- **Genetic factors**
- Mutations in *p16*, *BRAF*, or *MC1R* - Xeroderma pigmentosum
- **Immune suppression** (possible association, evidence debated)